



Module 20: Introduction to the ASA

Networking Security v1.0
(NETSEC)



Module Objectives

Module Title: Introduction to the ASA

Module Objective: Explain how the ASA operates as an advanced stateful firewall.

Topic Title	Topic Objective
ASA Solutions	Compare ASA solutions to other routing firewall technologies.
The ASA 5506-X with FirePOWER Services	Describe three ASA deployment scenarios.

20.1 ASA Solutions

ASA Firewall Models

The Cisco ASA with FirePOWER Services family of products provides dedicated firewall services in one device. These are next-generation firewall (NGFW) devices that deliver integrated threat defense across the entire attack continuum. They combine proven ASA firewalls with Sourcefire threat and advanced malware protection in a single device. These include Cisco Firepower 1000, Cisco Firepower 2100, Cisco Firepower 4100, Cisco Firepower 9300. All models provide advanced stateful firewall features and VPN functionality. The biggest difference between the models is the maximum traffic throughput handled by each model and the number and types of interfaces.

Cisco also supports the virtualization of computing infrastructure by taking advantage of the increased power availability of modern x86 servers. The Cisco Adaptive Security Virtual Appliance (ASAv) brings the power of ASA appliances to the virtual domain.

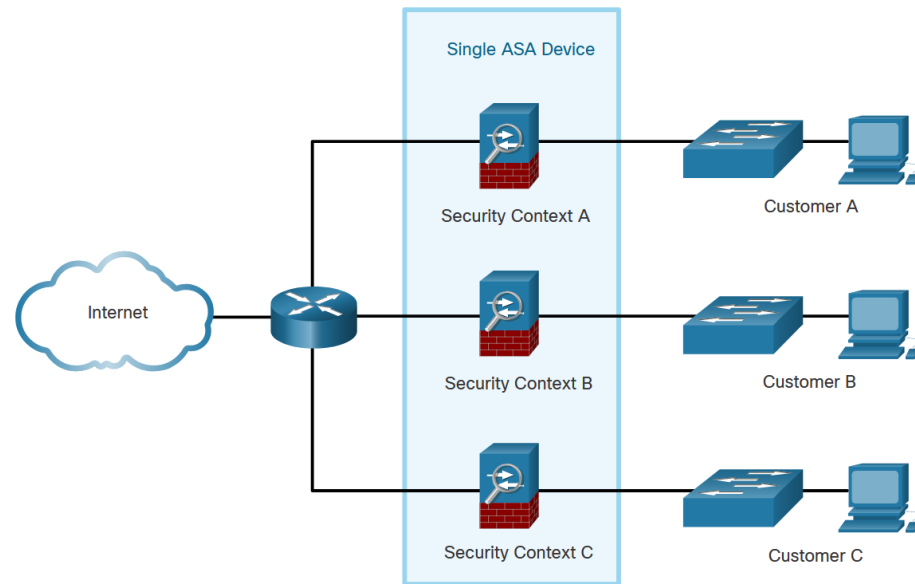
To provide a suitable fit for customer needs, Cisco ASAv is available in five models: Cisco ASAv5, Cisco ASAv10, Cisco ASAv30, Cisco ASAv50, Cisco ASAv100. The focus of this module will be on the ASA 5506-X which is designed for small business, branch office, and enterprise teleworker implementations.

Video - Cisco ASA Next-Generation Firewall Appliances

This video explains the Cisco ASA Next-Generation Firewall Appliances.

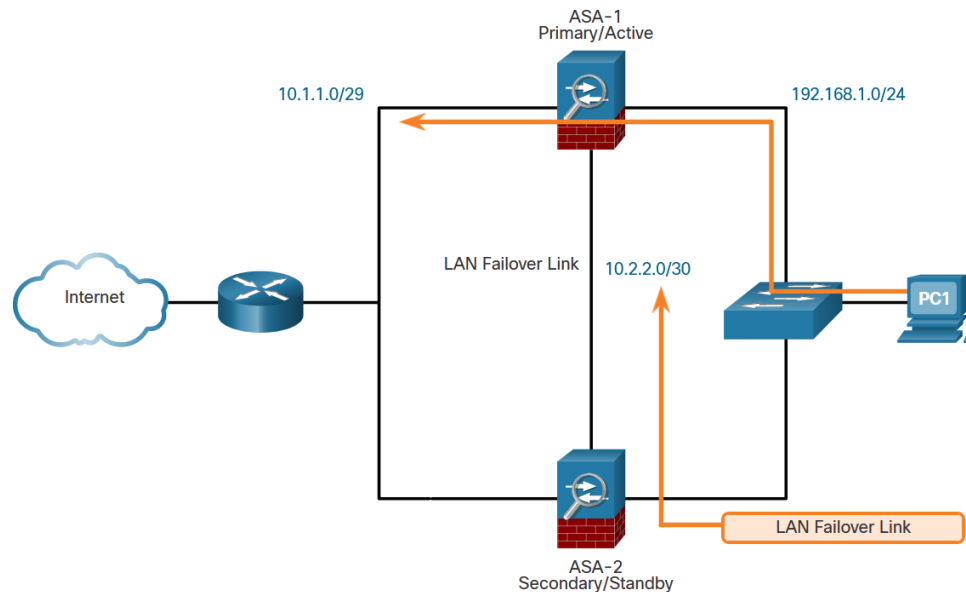
Advanced ASA Firewall Features

A single ASA can be partitioned into multiple virtual devices, as illustrated here. Each virtual device is called a security context. Each context is an independent device, with its own security policy, interfaces, and administrators. Multiple contexts are similar to having multiple standalone devices. Many features are supported in multiple context modes, including routing tables, firewall features, IPS, and management. Some features are not supported, including VPN and dynamic routing protocols.



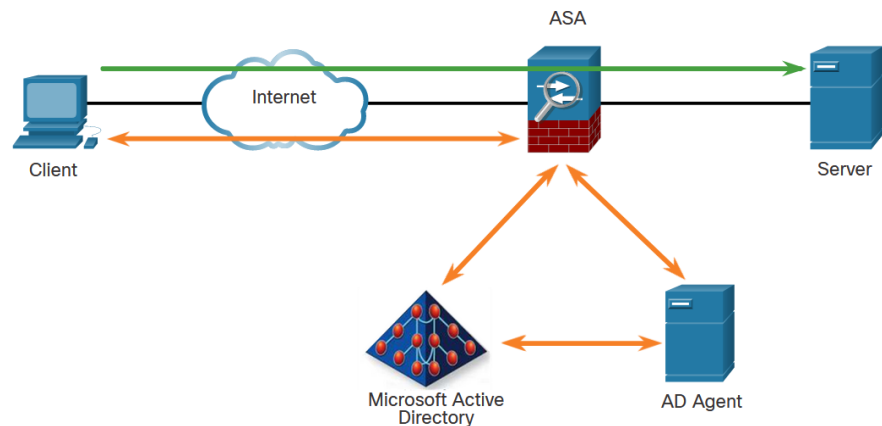
Advanced ASA Firewall Features (Cont.)

As shown here, two identical ASAs can be paired into an active / standby failover configuration to provide device redundancy. Both platforms must be identical in software, licensing, memory, and interfaces, including the Security Services Module (SSM). In the example, ASA-1 is the primary/active forwarding device and traffic leaving PC-1 takes the preferred path using ASA-1. ASA-1 and ASA-2 monitor each other using the LAN failover link. If ASA-1 fails, then ASA-2 would immediately assume the primary role and become active.



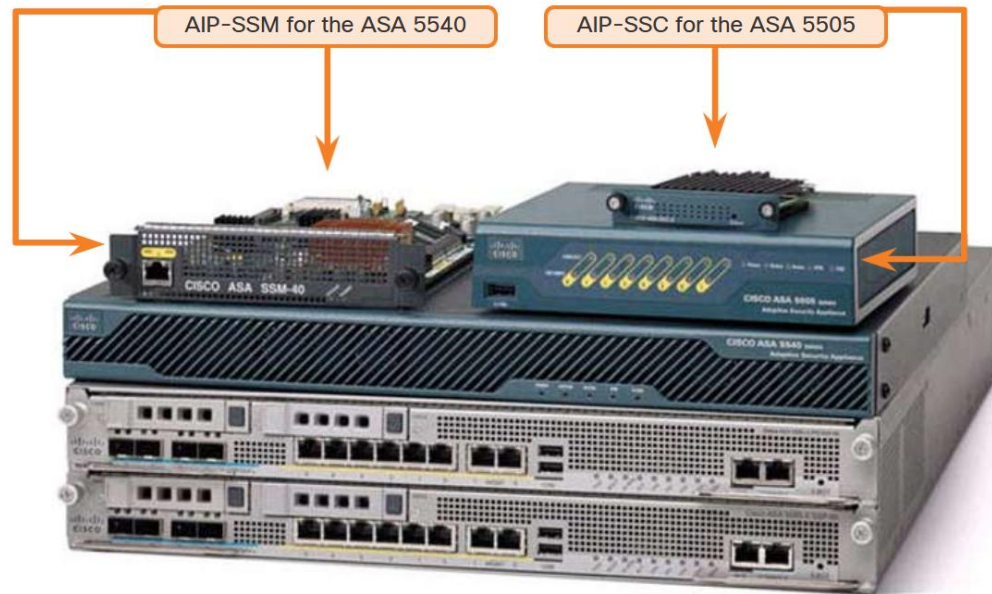
Advanced ASA Firewall Features (Cont.)

The ASA provides optional, granular access control based on an association of IP addresses to Windows Active Directory login information. For example, in the figure, when a client attempts to access the server resources, it must first be authenticated using the Microsoft Active Directory Identity-based firewall services. These services enhance the existing access control and security policy mechanisms by allowing users, or groups, to be specified in place of source IP addresses. Identity-based security policies can be interleaved without restriction between traditional IP address-based rules.



Advanced ASA Firewall Features (Cont.)

The ASA uses the Advanced Inspection and Prevention (AIP) modules. Antimalware capabilities can be deployed by integrating the Content Security and Control (CSC) module. The Cisco Advanced Inspection and Prevention Security Services Module (AIP-SSM) and Cisco Advanced Inspection and Prevention Security Services Card (AIP-SSC) deliver protection against tens of thousands of known exploits.



Cisco Firepower Series

Cisco has introduced a new line of next-generation firewalls (NGFW) which combines their proven firewall technology with Sourcefire advanced threat and malware detection capabilities.

These NGFWs consolidate multiple security layers into a single platform, eliminating the cost of buying and managing multiple solutions. This integrated approach combines best-in-class security technology with multilayer protection that is integrated into a single device.

The Cisco ASA 5500-X with FirePOWER Services devices are part of the new Cisco NGFWs. Designed for small to medium branch offices, the ASA 5500-X with FirePOWER Services merges the ASA 5500 stateful firewall features with some of the following advanced threat and malware detection capabilities:

- Next-generation IPS (NGIPS)
- Advanced Malware Protection (AMP)
- Application control and URL filtering

Video - Collect Firepower Threat Defense (FTD) Packet Captures with Firepower Management Center (FMC)

This video explains Firepower Threat Defense Packet Captures with Firepower Management Center.

Review of Firewalls in Network Design

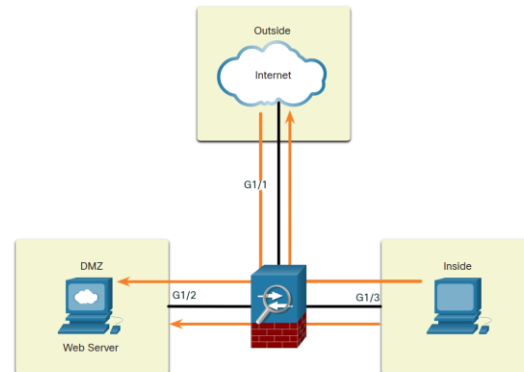
When discussing networks that are connected to a firewall, there are some general terms to consider:

- **Outside network** - The network/zone that is outside the protection of the firewall.
- **Inside network** - The network/zone that is protected and behind the firewall.
- **DMZ** - The demilitarized zone that allows both inside and outside users access to protected network resources.

Firewalls protect inside networks from unauthorized access by users who are on an outside network. They also protect inside network users from each other. For example, by creating zones, an administrator can keep the network that is hosting the accounting servers separate from other networks in an organization.

The figure illustrates how these zones interact for permitted traffic:

- Traffic originating from the inside network going to the outside network is permitted.
- Traffic originating from the inside network going to the DMZ network is permitted.
- Traffic originating from the outside network going to the DMZ network is selectively permitted.

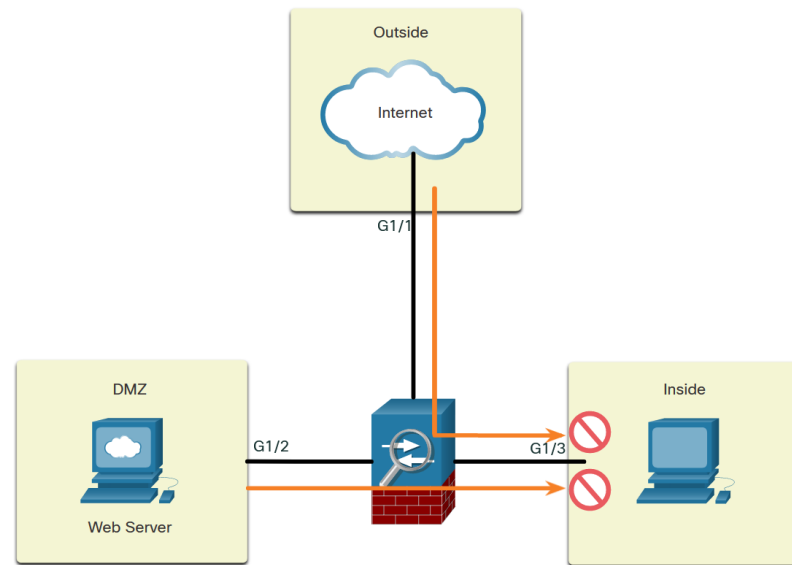


Review of Firewalls in Network Design (Cont.)

The figure illustrates how these zones interact for denied traffic:

- Traffic originating from the outside network going to the inside network is denied.
- Traffic originating from the DMZ network going to the inside network is denied.

Cisco ISRs can provide firewall features by using either the Zone-Based Policy Firewall (ZPF) or by using the older context-based access control (CBAC) feature.



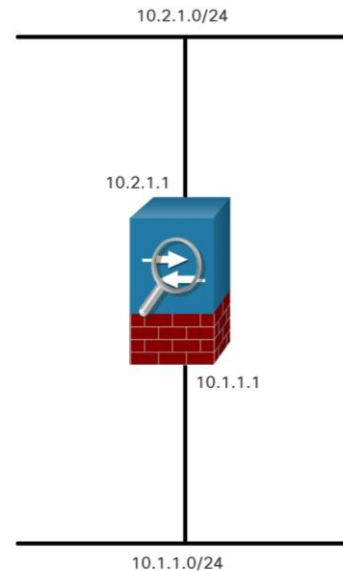
ASA Firewall Modes of Operation

There are two firewall interface modes of operation available on ASA devices: routed mode and transparent mode.

In routed mode, two or more interfaces separate Layer 3 networks, i.e. domains.

In the figure, the ASA is considered to be a router hop in the network and can perform NAT between connected networks. Routed mode supports multiple interfaces. Each interface is on a different subnet and requires an IP address on that subnet. The ASA applies policies to flows as they transit the firewall.

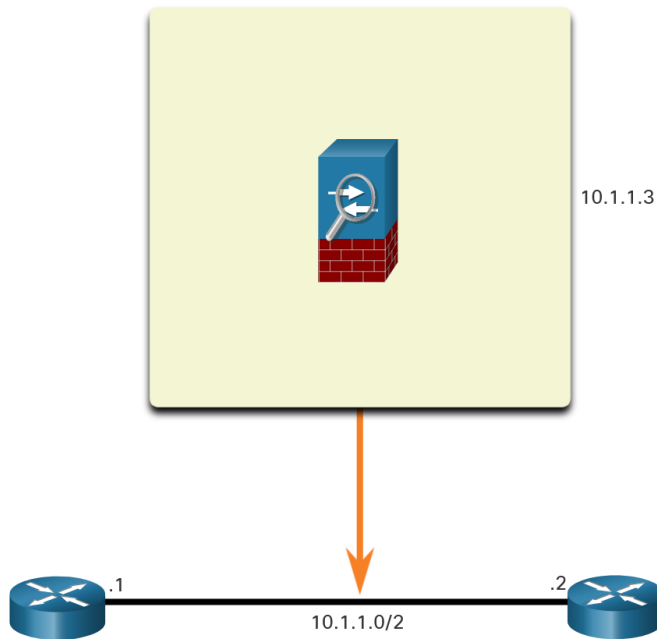
Routed Mode



ASA Firewall Modes of Operation (Cont.)

An ASA in transparent mode is often referred to as a “bump in the wire,” or a “stealth firewall” because the ASA functions like a Layer 2 device and is not considered a router hop. In the figure below, the ASA is only assigned an IP address on the local network for management purposes. This mode is useful to simplify a network configuration, or when the existing IP addressing cannot be altered. However, the drawbacks include no support for dynamic routing protocols, VPNs, QoS, or DHCP Relay.

Transparent Mode



ASA Licensing Requirements

A license specifies the options that are enabled on a given ASA. Most ASA appliances come pre-installed with either a Base license or a Security Plus license.

To provide more features to the ASA, additional time-based or optional licenses can be purchased.

To verify the license information on an ASA device, use the **show activation-key** command, as shown below, or the **show version** command.

```
NETSEC-ASA# show activation-key
```

```
Serial Number: JAD242301E6
```

```
Running Permanent Activation Key: 0x1e14e468 0x7c715e6b 0xcc71d1f4 0x9de81084 0x4e143eb6
```

```
Licensed features for this platform:
```

Maximum Physical Interfaces	: Unlimited	perpetual
Maximum VLANs	: 5	perpetual
Inside Hosts	: Unlimited	perpetual
Failover	: Disabled	perpetual
Encryption-DES	: Enabled	perpetual
Encryption-3DES-AES	: Enabled	perpetual
Carrier	: Disabled	perpetual
AnyConnect Premium Peers	: 2	perpetual
AnyConnect Essentials	: Disabled	perpetual
Other VPN Peers	: 10	perpetual
Total VPN Peers	: 12	perpetual
AnyConnect for Mobile	: Disabled	perpetual
AnyConnect for Cisco VPN Phone	: Disabled	perpetual
Advanced Endpoint Assessment	: Disabled	perpetual
Shared License	: Disabled	perpetual
Total TLS Proxy Sessions	: 2	perpetual
Botnet Traffic Filter	: Disabled	perpetual
Cluster	: Disabled	perpetual

```
This platform has a Base license.
```

```
The flash permanent activation key is the SAME as the running permanent key.
```

```
NETSEC-ASA#
```


20.2 The ASA 5506-X with FirePOWER Services

Overview of ASA 5506-X

The Cisco ASA 5506-X delivers a high-performance firewall, SSL VPN, IPsec VPN, and rich networking services in a modular, plug-and-play appliance. The default DRAM memory is 4 GB and the default internal flash memory is 8 GB. In a failover configuration, the two units must be identical models with the same hardware configuration, the same number and types of interfaces, and the same amount of RAM. Failover is available with the Security Plus license.

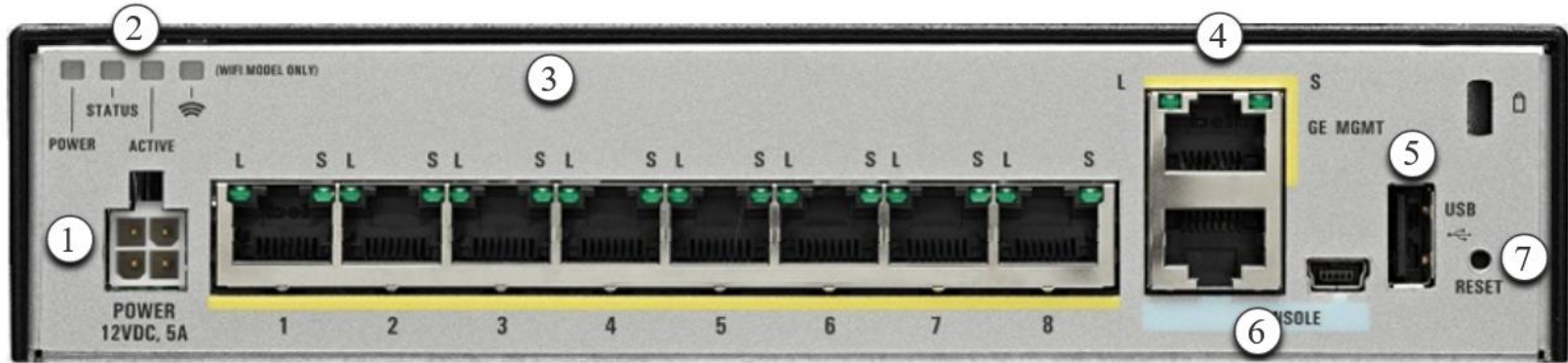
The figure illustrates the front panel of the ASA 5506-X.



The ASA 5506-X with FirePOWER Services

Overview of ASA 5506-X

The figure illustrates the back panel of the Cisco ASA 5506-X.



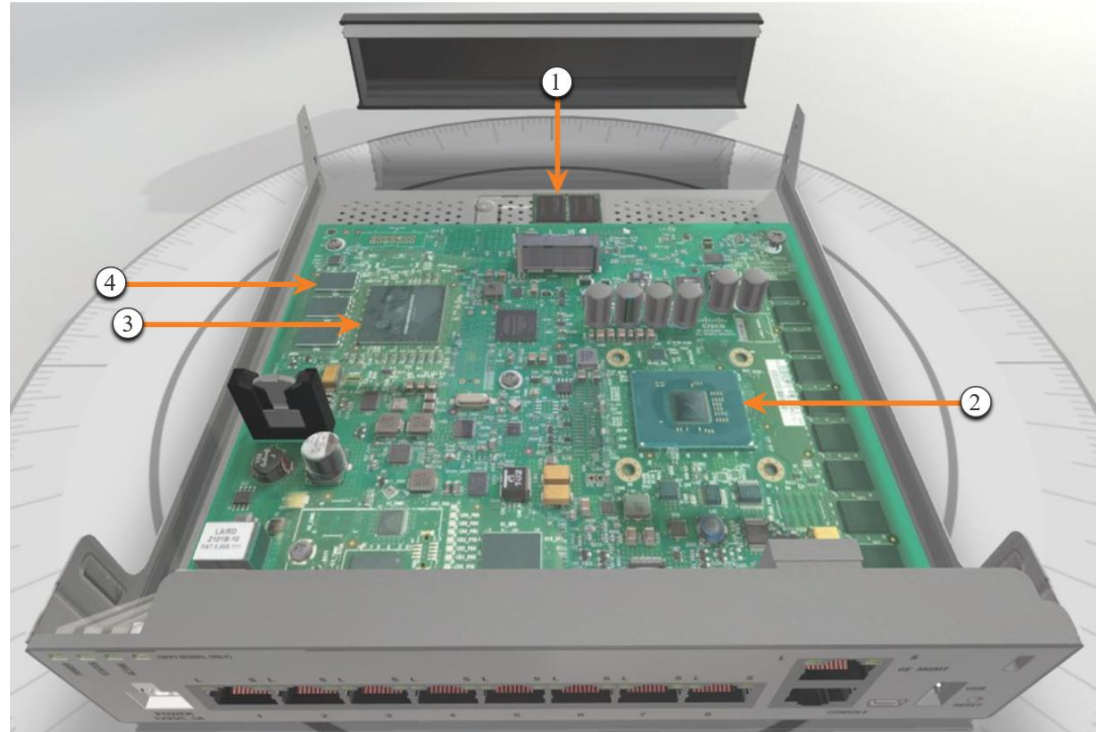
1. 12V power connector.
2. Four status LEDs for Power, Status, Active, and WLAN.
3. 8 - 1 Gigabit Ethernet interfaces. The L indicates link status and the S indicates connection speed.
4. Gigabit Ethernet management interface.
5. USB Type A port connection for external storage device.
6. RJ45 and mini-USB console ports.
7. Reset pin. When pressed for three seconds, the ASA resets to default new state on reboot. Flash is not erased.

The ASA 5506-X with FirePOWER Services

Overview of ASA 5506-X (Cont.)

The figure shows the inside components of the Cisco ASA 5506-X.

Unlike the ASA 5505, the ASA 5506-X does not use switchports. All interfaces are routed and require IP addresses.

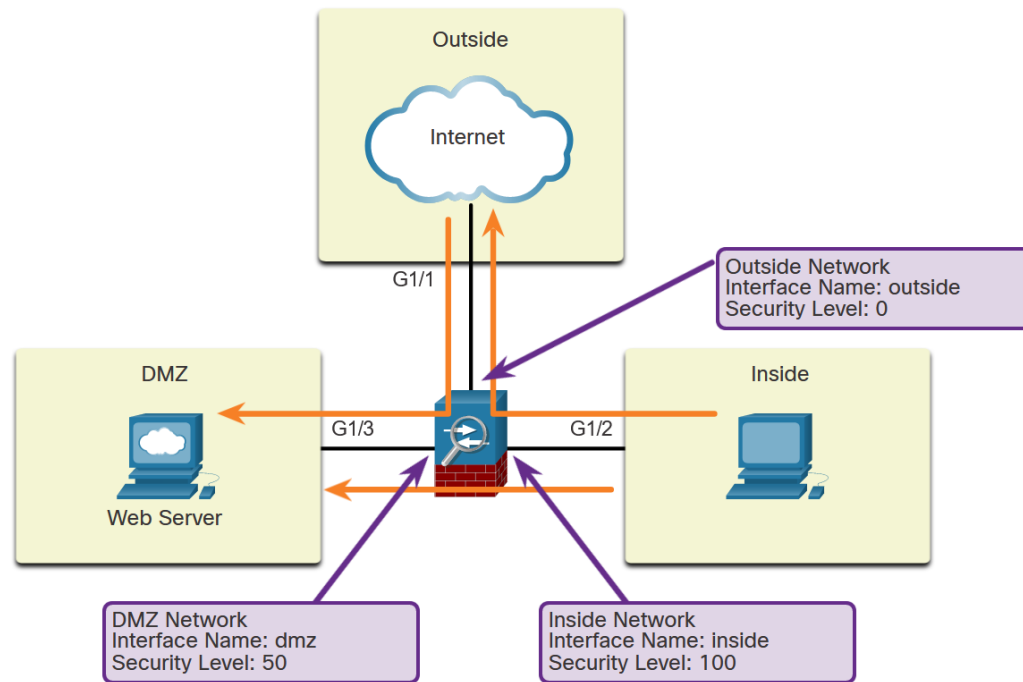


- 1. 50GB mSata solid state drive
- 2. Crypto accelerator CPU
- 3. Multicore 1.25 GHz CPU
- 4. 4GB DRAM

ASA Security Levels

Security levels define the level of trustworthiness of an interface. Each operational interface must have a name and a security level from 0 (lowest) to 100 (highest) assigned.

As shown in the figure, level 100 should be assigned to the most secure network, such as the inside network. Level 0 can be assigned to the outside network, which is connected to the internet. DMZs and other networks can be assigned a security level between 0 and 100



ASA Security Levels (Cont.)

Security levels help to control many aspects of network traffic as shown in the table below.

Aspect	Effect
Network Access	By default, there is an implicit permit from a higher security interface to a lower security interface (outbound). Hosts on the higher security interface can access hosts on a lower security interface. Multiple interfaces can be assigned the same security level. If communication is enabled for interfaces with the same security level, there is an implicit permit for traffic between the interfaces.
Inspection Engines	Some application inspection engines are dependent on the security level. When interfaces have the same security level, the ASA inspects traffic in either direction.
Application Filtering	HTTPS and FTP filtering applies only for outbound connections that are from a higher level to a lower level. If communication is enabled for interfaces with the same security level, traffic can be filtered in either direction.

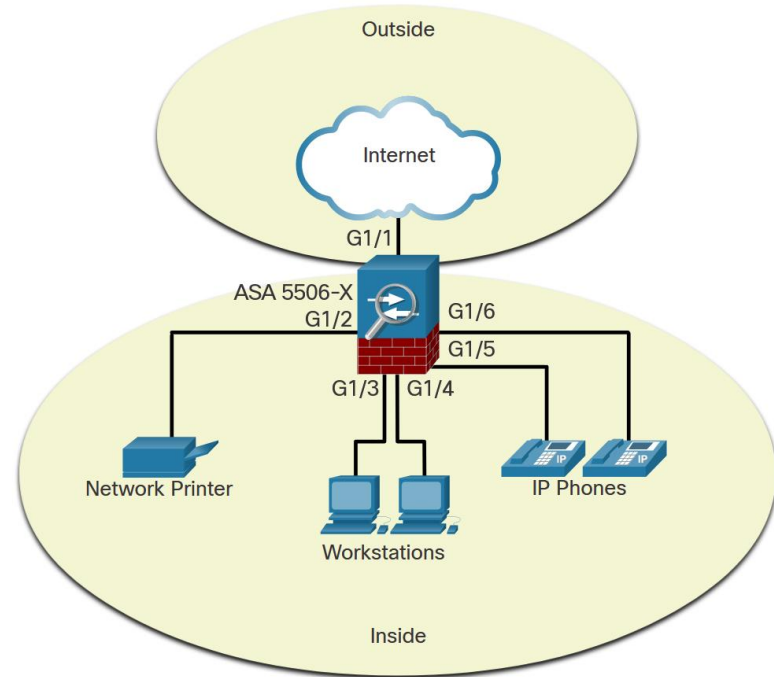
The ASA 5506-X with FirePOWER Services

ASA 5506-X Deployment Scenarios

The ASA 5506-X is commonly used as an edge security device. It connects a small business to an ISP device, such as a DSL or cable modem, for access to the internet. It can be deployed to interconnect and protect several workstations, network printers, and IP phones.

In a small branch, a common deployment would include an inside network (VLAN 1) with security level 100 and an outside network (VLAN 2) with security level 0, as shown in the figure.

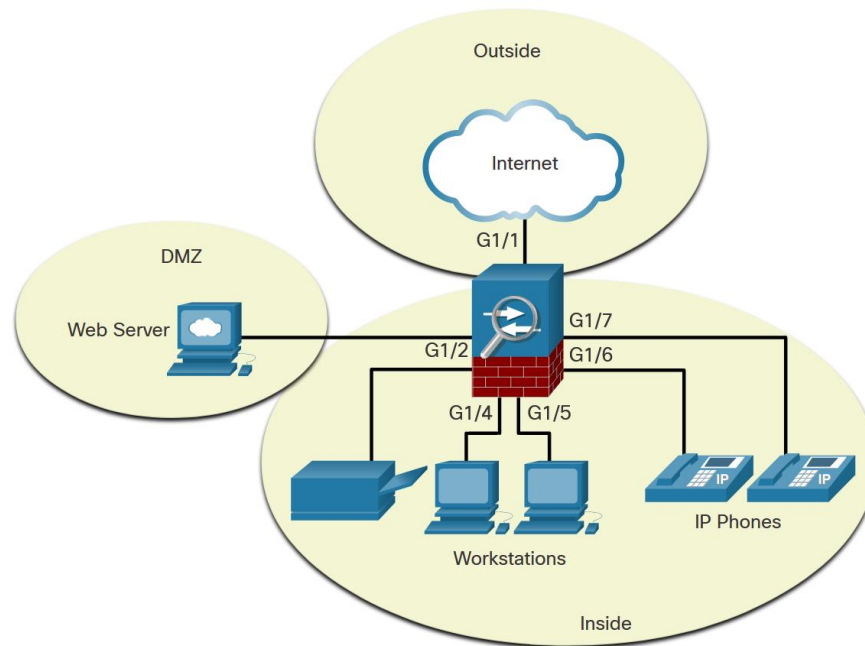
Small Branch



ASA 5506-X Deployment Scenarios (Cont.)

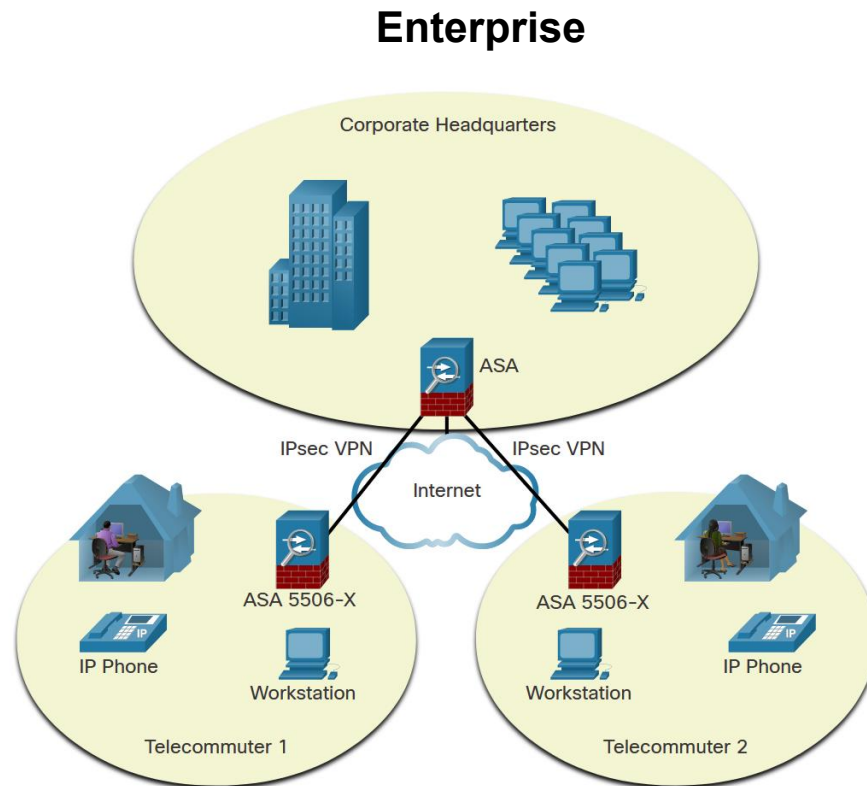
In the small business, shown here, the ASA 5505 can be deployed with two different protected network segments. One segment is the inside network (VLAN 1), which connects workstations and IP phones. The other segment is the DMZ (VLAN 3), which connects a company web server. The outside interface (VLAN 2) is used to connect to the internet.

Small Business



ASA 5506-X Deployment Scenarios (Cont.)

In an enterprise deployment, as shown here, the ASA 5505 can be used by telecommuters and home users to connect to a centralized location using a VPN.



20.3 Introduction to the ASA

Summary

What Did I Learn in this Module?

- The choice of ASA model depends on an organization's requirements, such as maximum throughput, maximum connections per second, and budget.
- When discussing networks connected to a firewall, terms to consider include outside network, inside network, and the DMZ.
- There are two firewall interface modes of operation available on ASA devices: routed mode and transparent mode.
- Advanced ASA firewall features include ASA virtualization, high availability with failover, identity firewall, and threat control and containment services.
- Most ASA appliances come pre-installed with either a Base license or a Security Plus license.
- The Cisco ASA 5506-X delivers a high-performance firewall, SSL VPN, IPsec VPN, and rich networking services in a plug-and-play appliance.
- The security level numbers range from 0 (untrustworthy) to 100 (very trustworthy).
- The ASA 5506-X is commonly used as an edge security device. It connects a small business to an ISP device, such as a DSL or cable modem, for access to the internet.

New Terms and Commands

- Security Services Module (SSM)
- Content Security and Control (CSC) module
- Advanced Inspection and Prevention Security Services Module (AIP-SSM)
- Advanced Inspection and Prevention Security Services Card (AIP-SSC)
- Next-generation IPS (NGIPS)
- Outside network
- Inside network
- context-based access control (CBAC)
- routed mode
- transparent mode
- **show activation-key**
- inspection engines
- application filtering

